

Database Management System

ECS 403

Lecture 15: Aggregate Functions, Null Values

By

Ms. Bapita Roy

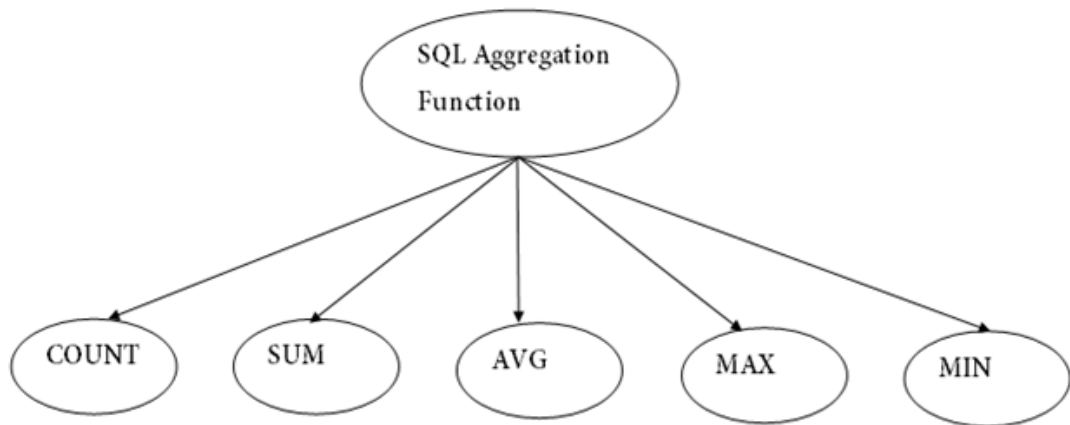
Head of the department

Electronics & Computer Science Engineering

Guru Nanak Institute of Technology

Contact no: 9231893138, Mail Id: head_ecs.gnit@gnit.ac.in

Types of SQL Aggregation Function



PRODUCT_MAST:

PRODUCT	COMPANY	QTY	RATE	COST
Item1	Com1	2	10	20
Item2	Com2	3	25	75
Item3	Com1	2	30	60
Item4	Com3	5	10	50
Item5	Com2	2	20	40
Item6	Cpm1	3	25	75
Item7	Com1	5	30	150
Item8	Com1	3	10	30
Item9	Com2	2	25	50
Item10	Com3	4	30	120

ST;

h WHERE

ST;

with DISTINCT

(DISTINCT COMPANY)

ST;

Example: COUNT() with GROUP BY

```
SELECT COMPANY, COUNT(*)  
FROM PRODUCT_MAST  
GROUP BY COMPANY;
```

Output:

Com1	5
Com2	3
Com3	2

Example: COUNT() with HAVING

```
SELECT COMPANY, COUNT(*)  
FROM PRODUCT_MAST  
GROUP BY COMPANY  
HAVING COUNT(*)>2;
```

Output:

Com1	5
Com2	3

2. SUM Function

Sum function is used to calculate the sum of all selected columns. It works on numeric fields only.

Syntax

```
SUM()  
or  
SUM( [ALL|DISTINCT] expression )
```

Example: SUM()

```
SELECT SUM(COST)  
FROM PRODUCT_MAST;
```

Output:

```
670
```

Example: SUM() with WHERE

```
SELECT SUM(COST)
FROM PRODUCT_MAST
WHERE QTY>3;
```

Output:

320

**Example: SUM() with GROUP BY**

```
SELECT SUM(COST)
FROM PRODUCT_MAST
WHERE QTY>3
GROUP BY COMPANY;
```

Output:

Com1 150

Com2 170



Example: SUM() with HAVING

```
SELECT COMPANY, SUM(COST)
FROM PRODUCT_MAST
GROUP BY COMPANY
HAVING SUM(COST) >= 170;
```

Output:

Com1	335
Com3	170

3. AVG function

The AVG function is used to calculate the average value of the numeric type. AVG function returns the average of all non-Null values.

Syntax

```
AVG()
or
AVG( [ALL|DISTINCT] expression )
```


Example:

```
SELECT AVG(COST)
FROM PRODUCT_MAST;
```

Output:

67.00

4. MAX Function

MAX function is used to find the maximum value of a certain column. This function determines the largest value of all selected values of a column.

Syntax

```
MAX()
or
MAX( [ALL|DISTINCT] expression )
```

Example:

```
SELECT MAX(RATE)
FROM PRODUCT_MAST;
```

30

5. MIN Function

MIN function is used to find the minimum value of a certain column. This function determines the smallest value of all selected values of a column.

Syntax

```
MIN()  
or  
MIN( [ALL|DISTINCT] expression )
```

Example:

```
SELECT MIN(RATE)  
FROM PRODUCT_MAST;
```

Output:

```
10
```

What is a NULL Value?

A field with a NULL value is a field with no value.

If a field in a table is optional, it is possible to insert a new record or update a record without adding a value to this field. Then, the field will be saved with a NULL value.

Note: A NULL value is different from a zero value or a field that contains spaces. A field with a NULL value is one that has been left blank during record creation

IS NULL Syntax

```
SELECT column_names  
FROM table_name  
WHERE column_name IS NULL;
```

IS NOT NULL Syntax

```
SELECT column_names  
FROM table_name  
WHERE column_name IS NOT NULL;
```

Consider the following CUSTOMERS table having the records as shown below.

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00
6	Komal	22	MP	
7	Muffy	24	Indore	

Now, following is the usage of the **IS NOT NULL** operator.

```
SQL> SELECT ID, NAME, AGE, ADDRESS, SALARY  
      FROM CUSTOMERS  
      WHERE SALARY IS NOT NULL;
```

This would produce the following result –

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000.00
2	Khilan	25	Delhi	1500.00
3	kaushik	23	Kota	2000.00
4	Chaitali	25	Mumbai	6500.00
5	Hardik	27	Bhopal	8500.00

Now, following is the usage of the **IS NULL** operator.

```
SQL> SELECT ID, NAME, AGE, ADDRESS, SALARY  
      FROM CUSTOMERS  
      WHERE SALARY IS NULL;
```

This would produce the following result –

ID	NAME	AGE	ADDRESS	SALARY
6	Komal	22	MP	
7	Muffy	24	Indore	

Thank You